

Cite this: *Chem. Commun.*, 2012, **48**, 11623–11625

www.rsc.org/chemcomm

COMMUNICATION

Medicinal gold compounds form tight adducts with the copper chaperone Atox-1: biological and pharmacological implications†

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Received 11th September 2012, Accepted 15th October 2012

DOI: 10.1039/c2cc36610j

Based on ESI-MS measurements, we show here that some representative cytotoxic gold(III) compounds produce stable adducts upon reaction with the copper chaperone Atox-1; notably, such adducts contain gold in the oxidation state +1. These findings are of interest to understand the intracellular metabolism of medicinal gold species and to develop new potent inhibitors of the copper trafficking system.

The so called “copper trafficking system” is primarily involved in the regulation of copper transport and homeostasis inside cells. Details of the copper trafficking system were recently disclosed.¹ This system consists of a few proteins that assist copper entrance inside cells and then promote its transfer and delivery to essential copper-dependent cellular proteins¹ (Fig. 1).

The sophisticated biochemical machinery of the *copper trafficking system* ensures a tight control of the intracellular concentrations of free copper ions that are potentially very deleterious to cells and are kept well below 10^{-15} M. *Atox-1* is an intracellular metallochaperone crucially involved in copper trafficking; structural details of *Atox-1* were clarified through a few recent studies.² Most *Atox-1* homologues are ~70 amino acid proteins containing a conserved CXXC motif for copper(I) binding in the vicinity of the N-terminus.

Previous investigations revealed that platinum based drugs do interact strongly with proteins belonging to the copper trafficking system; the latter, in turn, seems to play a key role in modulating cisplatin uptake.³ A strict linkage between the copper trafficking system and resistance to cisplatin was also pinpointed.⁴ More recent studies proved that cisplatin reacts eagerly with *Atox-1* forming stable derivatives.⁵ The structures of cisplatin derivatives of *Atox-1* have been solved through

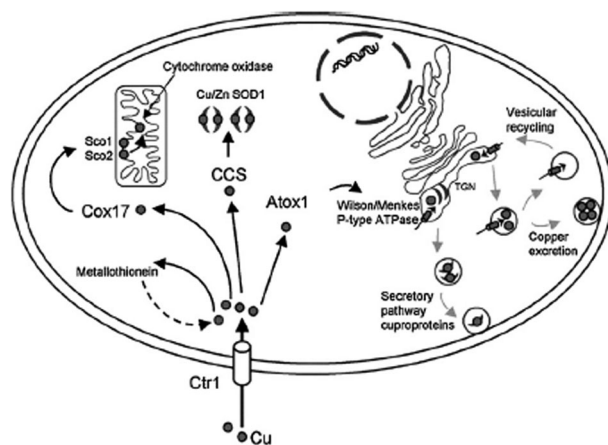


Fig. 1 Pathways of copper trafficking within a mammalian cell. Reproduced from ref. 5.

X-ray diffraction² or NMR measurements⁵ proving the interaction of the platinum center with the CXXC motif. Interestingly, Arnesano *et al.* showed that cisplatin binding to *Atox-1* comprises at least two steps: initially a cisplatin–*Atox-1* adduct is formed; afterward, protein dimerization and concomitant loss of amines from cisplatin take place.⁵ In turn, a study by Pernilla Wittung *et al.* analysed the reaction of cisplatin with *Atox-1* through various biophysical methods and demonstrated that copper(I) and platinum(II) ions may bind simultaneously to this protein to nearby, yet independent and not mutually exclusive, anchoring sites.⁶

During the last decade we were engaged in preparing and characterising a number of medicinal gold compounds as experimental cytotoxic and anticancer agents.⁷

On the ground of basic concepts of coordination chemistry and of established HSAB principles, we hypothesised that medicinal gold compounds, especially those in the oxidation state +1, might interact strongly with proteins of the *copper trafficking system*. Indeed, a previous study revealed that the copper trafficking system is substantially inhibited by “soft” d¹⁰ silver(I) ions, capable of binding *Atox-1*.⁸ Gold(I) ions – with a d¹⁰ electronic configuration – are even “softer” Lewis acids than silver(I) ions and, as such, should react eagerly with the copper(I) binding site of *Atox-1*; remarkably, a very recent study reported that copper transport proteins are involved in

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† Electronic supplementary information (ESI) available: Experimental section and an additional figure. See DOI: 10.1039/c2cc36610j

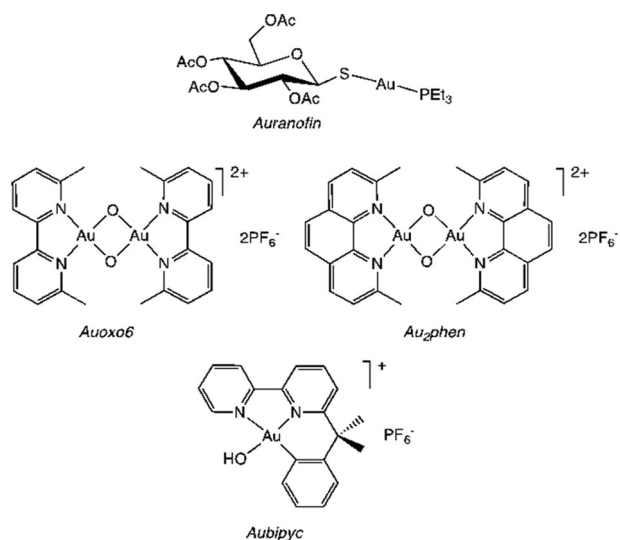


Chart 1 Structures of the tested compounds.

the uptake of some gold(I) carbene compounds.⁹ Interference with other biochemical pathways of transport and storage that regulate the homeostasis of divalent/trivalent transition metal ions¹⁰ cannot be ruled out at this stage.

Based on the above arguments, we decided to investigate the interactions of three representative cytotoxic gold(III) compounds developed in our laboratory – namely *Auoxo6*,¹¹ *Au₂phen*¹² and *Aubipyc*¹³ – and also of the clinically established gold(I) drug *auranofin* with human *Atox-1*. Schematic structures of the studied compounds are shown in Chart 1. The investigational gold(III) compounds, in particular *Auoxo6* and *Au₂phen*, have been reported to undergo rather facile reduction to gold(I), even under mildly reducing conditions, thus being an effective source of “soft” gold(I) ions. Typically, gold(III) reduction causes complete breakdown of the starting metal complex and release of its bulky bidentate organic ligands.

The interactions of the mentioned gold(III) compounds with *Atox-1* were analysed through ESI-MS as the latter method appears to be very suitable and informative to monitor the *Atox-1* protein in solution, as recently reported.⁵ For the ESI-MS measurements, *Atox-1* was dissolved in 25 mM tetramethylammonium acetate (TMeAmAc) buffer, containing DTT (with a 1 : 5 protein–DTT molar ratio) at physiological pH, and reacted with a stoichiometric amount of the above gold(III) compounds according to established experimental protocols (see ESI†). Samples were collected after 24 hours incubation and comparatively analysed through ESI-MS. Parallel studies were carried out with *auranofin* under identical solution conditions.

The resulting deconvoluted ESI-MS spectra are shown in Fig. 2. Formation of metal–protein adducts is clearly documented by the appearance of peaks of higher molecular mass than that of the native protein falling at ~7269 Da. The new peaks may be straightforwardly assigned to protein binding of gold containing molecular fragments. In the case of the three gold(III) compounds, the same major adduct was invariably formed (with a peak falling at ~7465 Da) matching protein binding of a single gold(I) ion. Formation of this adduct implies disruption of the starting gold(III) complex, loss of

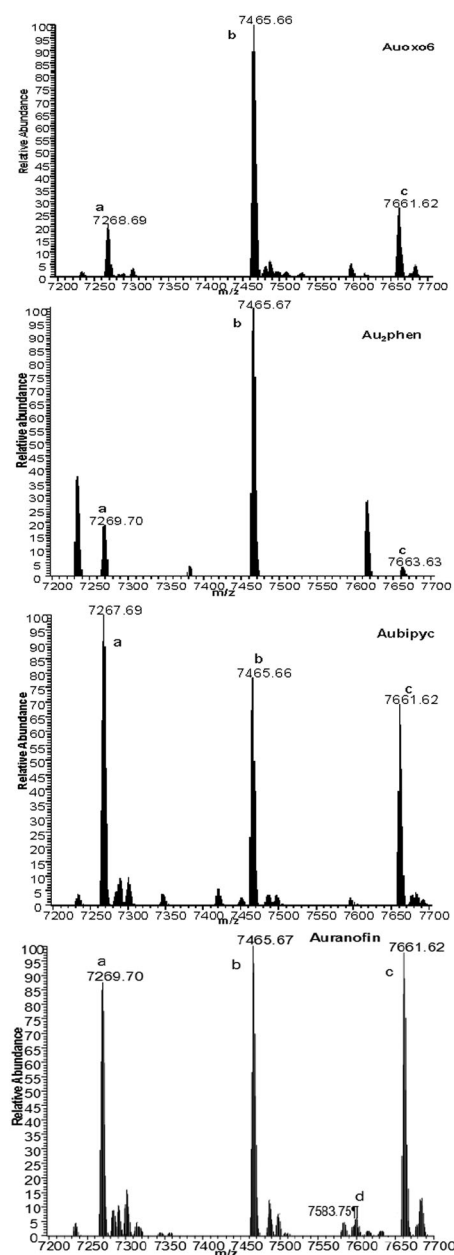


Fig. 2 LTQ-Orbitrap ESI-MS spectra of *Atox-1* (a) treated with gold compounds. *Atox-1* adducts are formed with: 1 Au⁺ (b) or 2 Au⁺ (c) bound to the protein; in the case of auranofin there is an additional adduct with a Au(PEt₃)⁺ fragment (d). Incubation time 24 h. Theoretical values for the main peaks (a–d) are reported in Table S1 (ESI†).

ligands and reduction of gold(III) to gold(I). Auranofin gave the same adduct as its major reaction product; however, the weak peak observed at ~7599 Da well matches an adduct bearing the Au(PEt₃)⁺ fragment.

In all cases the monometalated derivative resulted to be the predominant adduct; however, a second intense peak was invariably observed at ~7661 Da that corresponds to protein binding of two gold(I) ions. In two cases *i.e.* *Aubipyc* and *Auranofin* this latter peak manifests an intensity absolutely comparable to that of the monometalated adduct, implying that adducts containing two metal centers may be formed in large amounts.

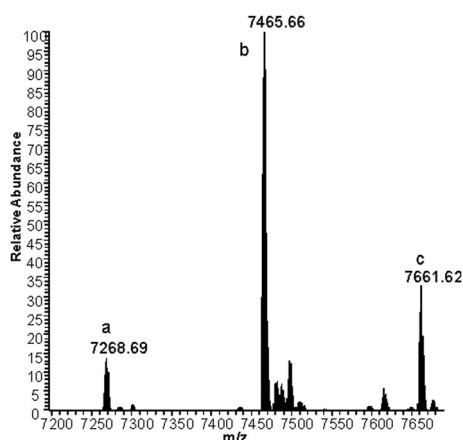


Fig. 3 LTQ-Orbitrap ESI-MS spectrum of *Atox*-treated with *Aubipyc* after 72 h incubation; other conditions as in Fig. 2.

The stability of the obtained gold-*Atox-1* adducts in solution, at room temperature, was further assessed by recording ESI-MS spectra at regular time intervals over 72 hours. An appreciable stability of gold-*Atox-1* adducts could be documented in selected cases. The behaviour of *Aubipyc* is very representative (Fig. 3). Indeed, its gold-protein adducts turned out to be fairly stable with time over 72 hours; yet, the total amount of gold adducts increased significantly with time as it emerges from comparative analysis of peak intensities at 24 hours (Fig. 2) versus 72 hours (Fig. 3). In contrast, the adducts formed with *Auoxo6* and *Au₂phen* manifested a lower stability over incubation times longer than 24 h showing evident signs of adduct and protein degradation (data not shown).

The above reported results well document that *Atox-1* can form tight derivatives with gold(I) ions implying that the resulting metal-protein interactions are strong. To better elucidate this point competition studies were carried out with copper. More precisely, *Atox-1* was preincubated with copper(II) ions in the presence of DTT, according to previous preparative procedures (see ESI,† Fig. S1); then, gold compounds were added. ESI-MS spectra were taken for the final samples after 12 hour incubation with gold compounds. Some representative ESI-MS data for these competition experiments are reported in the ESI.† Remarkably, large amounts of gold-*Atox-1* derivatives were formed while copper-*Atox-1* derivatives are hardly seen. Again, these results point out that gold ions apparently form far stronger *Atox-1* complexes than copper ions (Fig. S1, ESI†).

In conclusion, we have explored here whether a few representative cytotoxic gold compounds might manifest high affinity and form stable adducts with the copper chaperone *Atox-1*. Notably, large metal-protein affinities and relevant adduct formation were unambiguously proved through ESI-MS analysis; the nature of the adducts and their metal-protein stoichiometries was elucidated at the molecular level.

Based on the above ESI-MS results, some general trends may be proposed for the investigated reactions. Under the applied experimental conditions – i.e. the reducing environment provided by DTT – the gold(III) centers are invariably reduced to gold(I) with simultaneous loss of the starting ligands. In turn, the resulting gold(I) species bind *Atox-1* strongly. Two major gold binding sites were disclosed on the protein that may well correspond to previously described

metal binding sites. Competition studies with copper were also carried out showing that gold-*Atox-1* adducts are far more stable than copper adducts.

In view of these findings, it may be suggested that gold compounds will interfere importantly with the intracellular copper trafficking system. This observation contains some relevant pharmacological implications: (i) it is likely that gold based medicinal compounds may exploit the copper trafficking system to enter cells and to be delivered to a number of cellular sites; *this kind of processing, however, causes disruption of the original gold(III) species and gold(III) reduction to gold(I)*; (ii) it can be hypothesised that gold compounds may drastically inhibit the copper trafficking system and be effective in modulating its efficiency; (iii) such effects might be relevant even for a modulation of cisplatin activity and for the control of its uptake.

Thus, the present results strongly warrant that gold compounds are tested *in vitro* on appropriate cellular systems to ascertain whether the predicted effects truly occur. Moreover, our study further stresses the importance of ESI-MS methods to elucidate adduct formation between metallo drugs and proteins and delineate the respective mechanisms of interaction.

Generous financial support from AIRC (IG-12085) and from Beneficentia Stiftung (Vaduz, Liechtenstein) is gratefully acknowledged.

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